



Course: Electronic Circuit Design
 Program: BS(Electrical Engineering)
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 Section: ALL
 Exam: Final

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 Weight: 50%
 Page(s): 3
 Questions: 5

Q

With the emitter used as a common terminal between input and output, Fig. 1 shows a transistor amplifier known as a **common-emitter** or **grounded-emitter** circuit. The circuit parameters are: $R_s = 5 \text{ k}\Omega$, $r_\pi = 2.5 \text{ k}\Omega$, $g_m = 40 \text{ mA/V}$, $r_o = 100 \text{ k}\Omega$, and $R_L = 5 \text{ k}\Omega$.

- Derive an expression for the voltage gain v_o/v_i .
- Evaluate the magnitude of gain.
- Evaluate the magnitude of gain if the effect of r_o is neglected.
- An alternative model for the transistor in which the trans-conductance amplifier is replaced by a current amplifier is shown in Fig. 2. Evaluate the short-circuit current gain β .

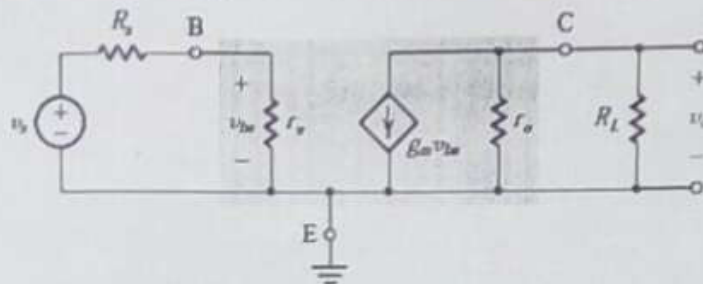


Fig. 1

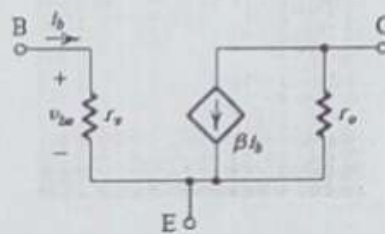


Fig. 2

Q2 An internally compensated op-amp has an open-loop voltage gain of 80 dB. The corner frequency occurs at 6 Hz. [20]

- Find the unity gain frequency (0 dB frequency).
- If the same op-amp is now connected in a closed-loop to form an inverting amplifier with a closed-loop gain of $G = -9 \text{ V/V}$. Find the corner frequency for this closed-loop amplifier.

b) Design a circuit using only one op-amp that provides output signal, given by Eq. 1, for an input signal of V_i . Only $100.0 \text{ k}\Omega$ resistors are to be used in the circuit design.

$$V_o = -V_i - 0.01 \frac{dV_i}{dt} \quad (1)$$

- Clearly construct the circuit schematic.
- Find the values of all the components used in the circuit.

- Q3 The 555 timer circuit shown in Fig. 3 is an astable multi-vibrator. [20]
 a) Write down the expressions for the time-period and the duty-cycle of the output waveform if a diode is connected anti-parallel to R_B . CLO 04
 b) Hence, design the values of R_A and R_B so that the circuit can generate a perfect square-wave signal (50% duty-cycle) having a 1000 Hz frequency. Use a $0.1 \mu\text{F}$ capacitor in the design. PLO 1.1
 c) Construct the diagram of an alternative circuit that uses only one op-amp, instead of 555 timer, to generate a perfect square wave of 1000 Hz using a capacitor of $0.1 \mu\text{F}$. Find the unknown resistances if $\beta = 0.5$. The maximum value of resistance allowed in this circuit is $1.0 \text{ M}\Omega$.

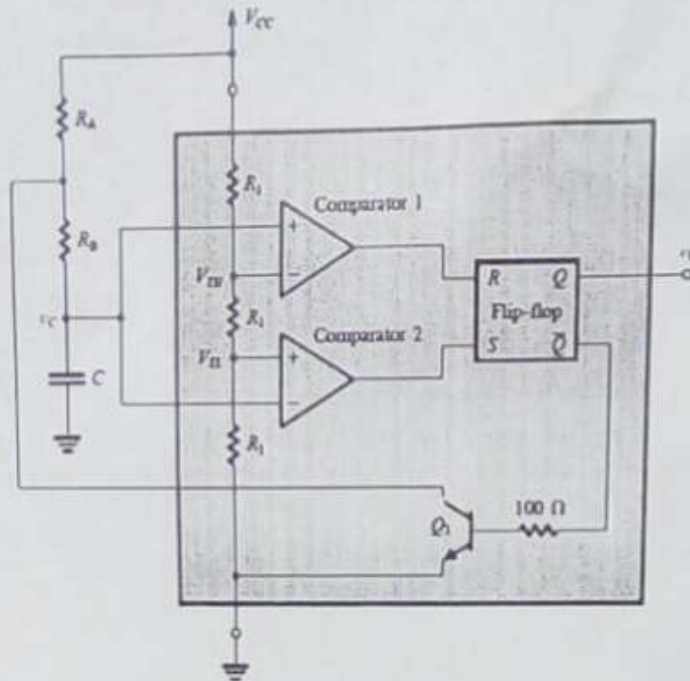


Fig. 3

- Q4 The circuit shown in Fig. 4 provides transfer characteristics with $\pm 7.5 \text{ V}$ levels at the output v_o . [20]
 When $v_i = 0 \text{ V}$, a current of 0.5 mA flows in the feedback resistor (R_2) and a current of 1.0 mA flows through the zener-diodes. Assume that the output saturation levels of the op-amp are $\pm 12.0 \text{ V}$, the forward-drop voltage of the zener-diode is 0.7 V , and the value of R_1 is equal to R_2 . CLO 04
 a) Specify the zener-voltages of the zener diodes. PLO 1.1
 b) Find the values of all resistors.

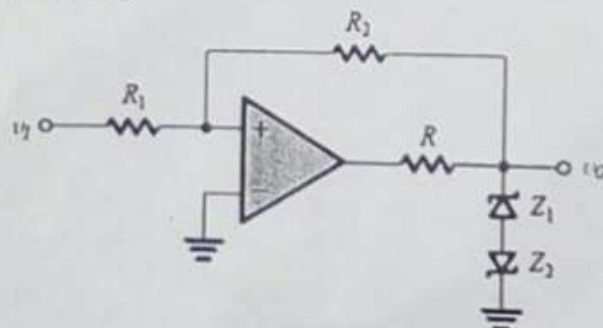


Fig. 4

- Q5 Clearly draw the CMOS implementation of $Y = AB(C+D)$ using: [20]
 a) NAND and NOR gates (draw the combinational logic circuit diagram as well)
 b) Post-inversion technique
 Do not use the pre-inversion technique! CLO 05